h30s — predicting the 30-second mid-price move on a 5-second grid

IEOR 4511 capstone · h30s group · status report generated 2026-11-06 from the registered runs (code: *scripts/h30s_report.py*)

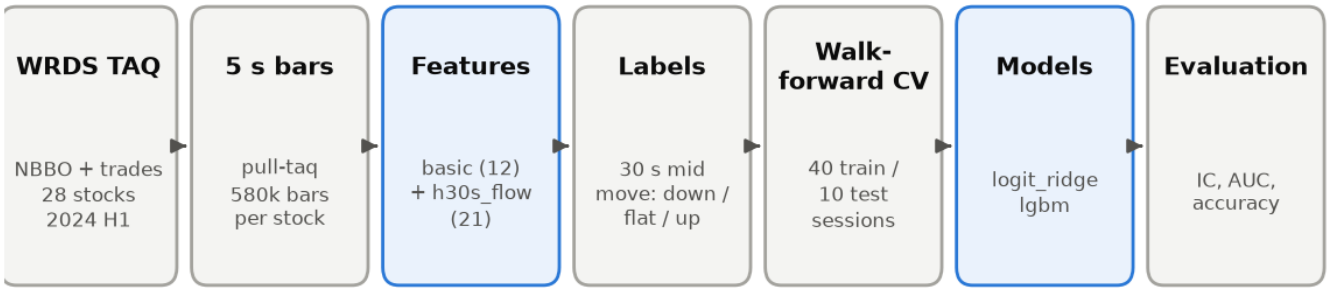
Summary

- **Task.** Predict whether the NBBO mid price of a stock is down, flat or up 30 seconds ahead, from level-1 quote data on 5-second bars, using the team framework (*hftf ml*).
- **Data.** WRDS TAQ, 28 large-cap S&P 500 stocks, 2024-01-02 to 2024-06-28 (124 sessions, 580,320 bars per stock). Walk-forward evaluation on 9 test windows: 10.4 M out-of-sample predictions (15.4 M usable rows in total).
- **Result.** There is real, consistent short-horizon predictability. The best model (LightGBM on the shipped features plus our order-flow features) reaches IC 0.099, AUC 0.638 and 46.9% accuracy, against 37.4% for always predicting the most common class. It beats the starting model in 9 of 9 test windows and in 27 of 28 stocks.
- **What drives it.** Level-1 queue imbalance (bid vs ask size) and recent order flow, and the tick regime: low-priced stocks, where one cent is a large step relative to the price, are far more predictable (mean IC 0.20 for the 8 stocks priced ≤ $50 vs 0.07 for the rest).
- **How big it is.** Small. The strongest decile of signals predicts a move of about 0.59 bps, below the average half-spread of 1.36 bps, so the signal does not pay for crossing the spread on its own; it is useful for timing and passive execution.
- **Look-ahead.** Labels never cross sessions, features use only past bars (checked automatically), CV is purged and embargoed walk-forward; all 367 framework tests pass (section 6).

Model (features)	Accuracy	Balanced acc.	AUC up	IC	IC t-stat	R^2 vs zero
zero (most common class)	37.4%	33.3%	–	–	–	–
last sign (momentum)	38.7%	38.6%	0.518	-0.010	–	–
logit_ridge · basic	45.0%	42.9%	0.616	0.091	125	0.8%
logit_ridge · basic + flow	46.1%	44.3%	0.631	0.094	112	0.9%
lgbm · basic + flow	46.9%	45.5%	0.638	0.099	145	1.0%

Pooled over 10,376,282 test rows in 84 test sessions. IC = rank correlation between the model's up-minus-down probability and the realized 30 s log return; the t-stat uses the per-session ICs. Classes in the test data: down 31.0%, flat 37.4%, up 31.6%.

1. What the pipeline does



Blue boxes: added by the h30s group (h30s_flow features, lgbm model). Grey: shared framework.

- **Bars.** *hftf data pull-taq* aggregates TAQ NBBO quotes (and trades) on the WRDS server into 5-second bars labelled at the bar end: last bid/ask and sizes, mid, microprice, time-weighted mid, quote count. 336 stock-months, 0 skipped, about 2 h 45 min, 2.6 GB.
- **Universe.** Our 28 pilot large caps (AAPL, MSFT, NVDA, ...). GOOG stands in for GOOGL and BRK.B is missing: *pull-taq* only reads symbols without a TAQ suffix (GOOGL is stored as GOOG + L, BRK.B as BRK + B).
- **Label.** The log change of the mid over the next 30 s (6 bars), classified down / flat / up with a 1 bp dead zone. Labels are computed within a session only; the first and last minute of each session and stale targets are dropped.
- **Cross-validation.** Purged, embargoed walk-forward by session: train on 40 sessions, test on the next 10, roll forward; training rows whose label overlaps the test window are removed. This gives 9 test windows covering 84 sessions (2024-02-29 to 2024-06-28).
- **Models.** *logit_ridge* (framework baseline: logistic regression for the direction, ridge for the return) and *lgbm* (ours). Each run also scores the *zero* and *last_sign* baselines on the same rows.

2. What the h30s group added

2.1 Data and two framework fixes

- Pulled the TAQ bars and pointed the h30s config at them (*horizons/h30s/config.yaml*, *symbols_pilot28.txt*).
- **Fixed a bug that affects every group** (*ml/datasets.load_bars*): the data cache keeps one growing file per stock, so on a multi-month pull every monthly manifest entry also contained the earlier months and January was loaded six times (2.0 M rows instead of 580 k per stock; 96% of rows were then dropped as bar gaps). Each entry is now sliced to its own dates and partitions are aligned by column name. Regression test added.
- Relaxed one shared test that required the model registry to equal the five shipped models exactly, which breaks as soon as a group registers its own model.

2.2 Order-flow feature set h30s_flow (21 features)

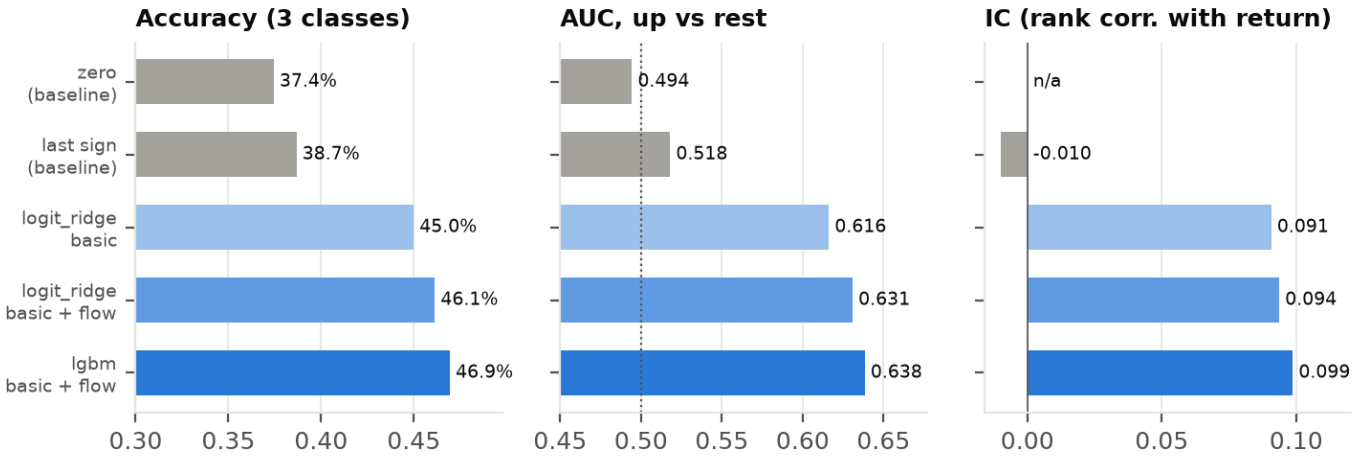
Group	Features	Idea
Order-flow imbalance	ofi_1, ofi_2, ofi_6, ofi_12, ofi_60	Cont–Kukanov–Stoikov OFI between consecutive bar-end NBBO snapshots, summed over 5 s ... 5 min and divided by average depth: net pressure from queue growth/depletion and price moves.
Imbalance persistence	imb_mean_6/12/60, imb_change_1/6	Is the bid/ask size imbalance building up or fading?
Tick regime	spread_ticks, one_tick_share, tick_bps, imb_x_one_tick	Spread in ticks, share of recent bars at a one-tick spread, tick size relative to price, and imbalance × one-tick share: imbalance matters most when the spread is pinned at one tick.
Microprice / intra-bar	microprice_ticks, twap_dev_bps, range_bps, quotes_rel	Microprice tilt in ticks, last mid vs the bar's time-weighted mid, mid range in the bar, quote intensity relative to its recent level.

The shipped *basic* set (12 features) has mid returns over 1/5/10 bars, spread, queue imbalance, microprice tilt in bps, quote count, a stale flag, time since the open and three volatility measures. Every new feature is a non-negative shift or a trailing window inside the session, and passes the framework's causality check.

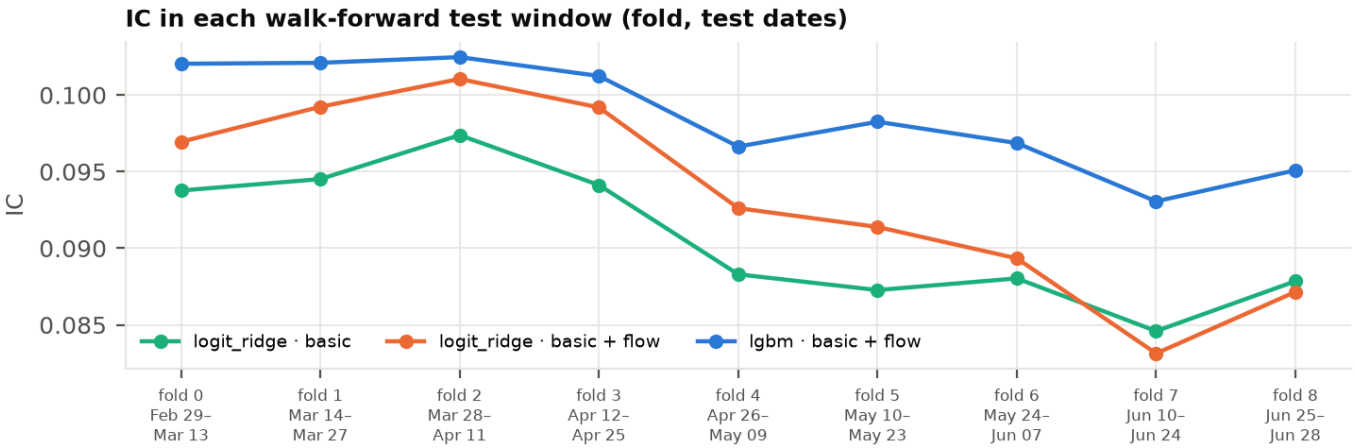
2.3 LightGBM model lgbm

Implements what the shared *gbm* stub describes, in our own file (*ml/models/h30s_lgbm.py*; the stub is untouched): one gradient-boosted booster for the direction and one for the signed return, using the framework's sample weights, seeded. The number of trees is chosen by early stopping on held-out blocks *inside the training window* (every 10th block of 20 000 rows, so validation rows span all stocks and dates); the test window is never used. On the last fold the direction booster used its full budget of 500 trees (early stopping did not trigger), so a larger budget or learning rate may still add a little (section 7).

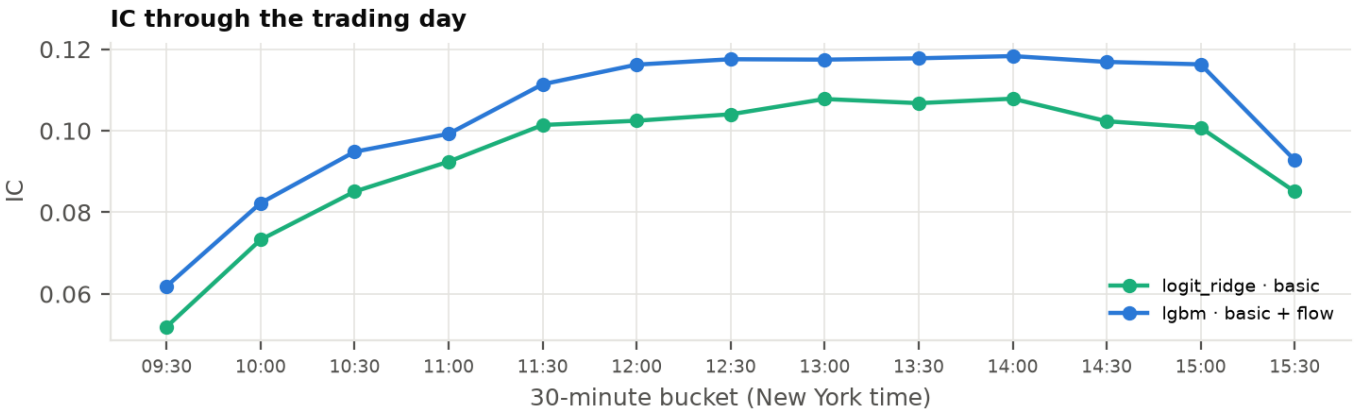
3. How the model performs



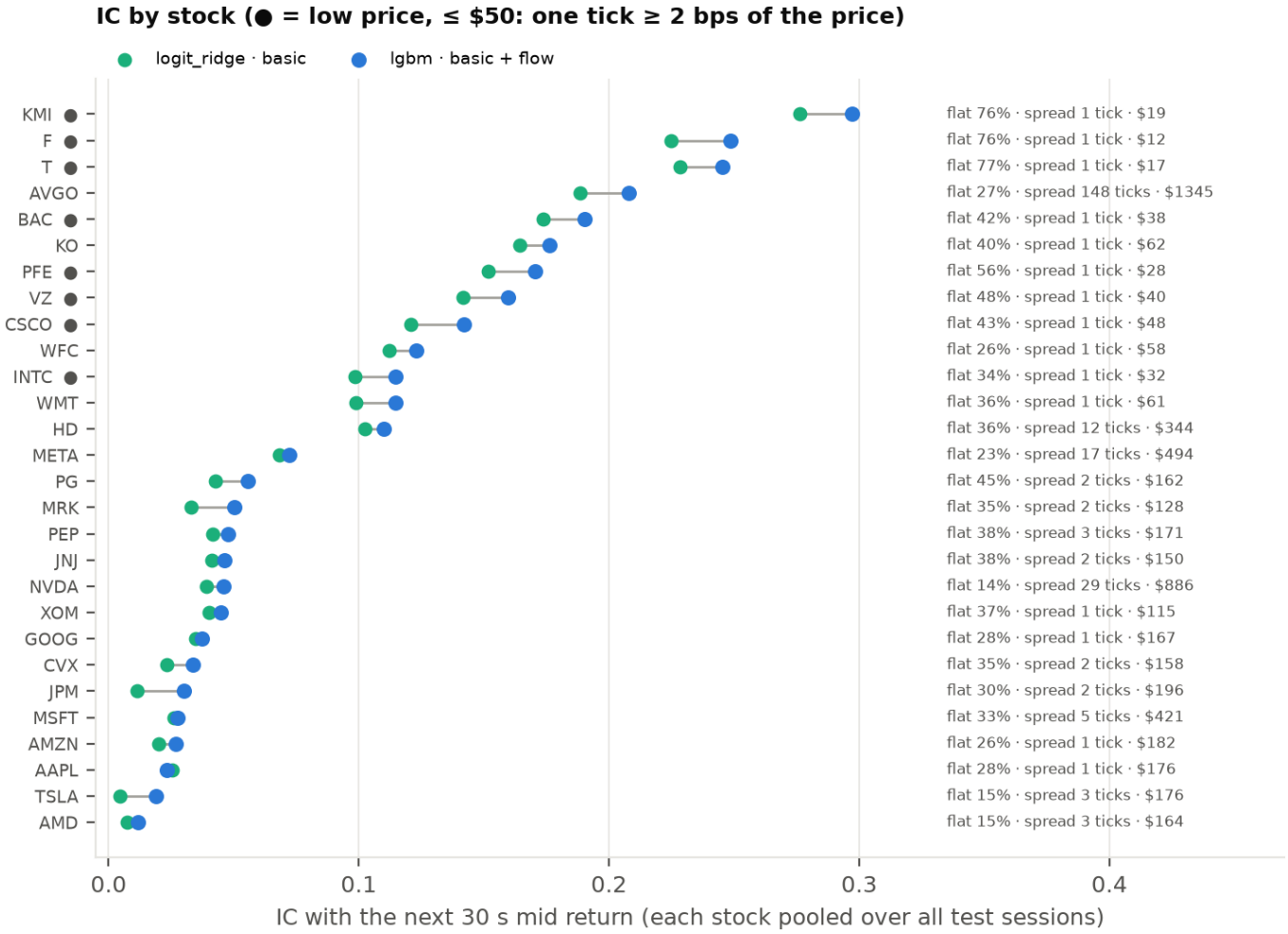
All runs are scored on the same 10.4 M test rows. Grey: framework baselines; blue shades: logit_ridge with the shipped features, logit_ridge with our order-flow features added, and LightGBM with both.



The improvement is not driven by one period: LightGBM with order flow has the highest IC in 9 of 9 test windows; adding the features to the linear model helps in 7 of 9.



Predictability is lowest right after the open, when the book is still forming, and highest from late morning through the afternoon.



IC rises for 27 of 28 stocks. Low-priced stocks (●, price ≤ $50, so one cent is ≥ 2 bps and the mid is often unchanged for 30 s) are far more predictable: mean IC 0.177 → 0.196; the other stocks 0.056 → 0.065. Many high-priced stocks also quote a one-tick spread (AAPL, AMZN, GOOG), but there a tick is under 1 bp. 'flat' = share of test rows with no move beyond the dead zone; spread and price are medians over the test rows.

4. What contributes to the result

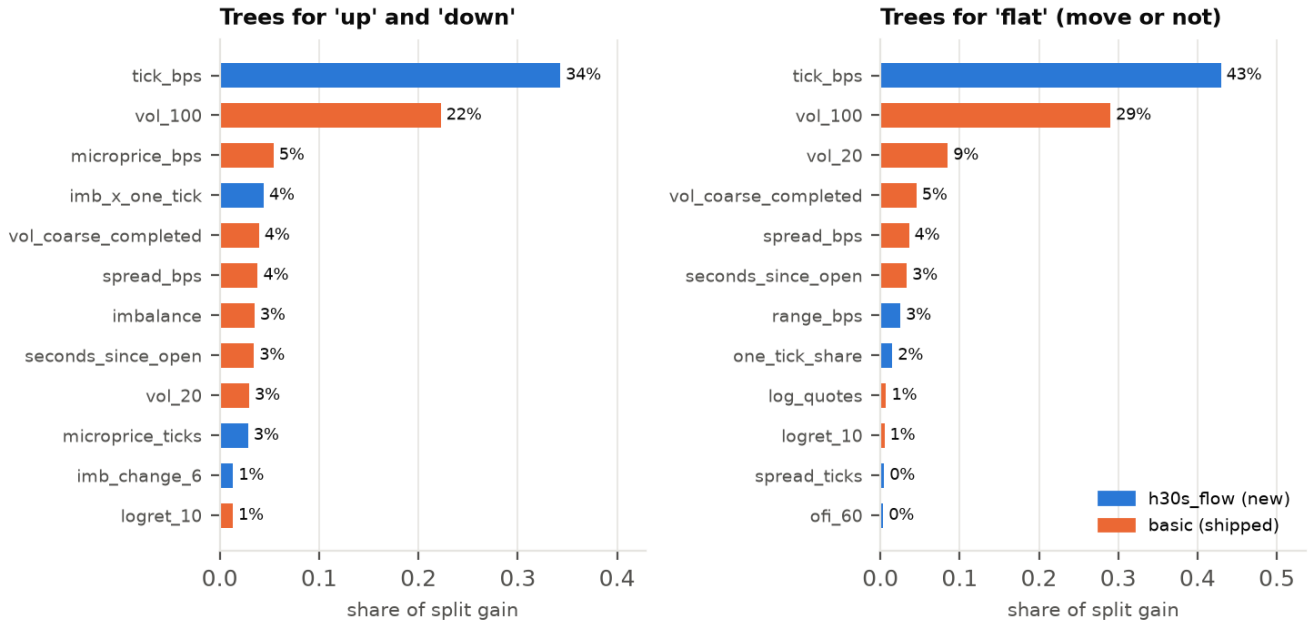
4.1 Step by step

Change	IC	Δ IC	Accuracy	AUC up
logit_ridge, shipped features	0.0907	–	45.0%	0.616
+ h30s_flow features (same linear model)	0.0937	+0.0030	46.1%	0.631
+ LightGBM instead of logit_ridge	0.0988	+0.0050	46.9%	0.638

About 37% of the IC gain comes from the new features and 63% from the nonlinear model. The model matters because the value of the imbalance depends on the tick regime, an interaction a linear model only sees through the hand-made imb_x_one_tick term.

4.2 Which inputs the model uses

LightGBM gain importance, refit on fold 8's training rows only (4.9 M rows)



Gain importance from a diagnostic refit of the lgbm direction booster on fold 8's training sessions only (same settings as the run). LightGBM grows one tree per class per round, so the gain can be split exactly into the trees that score 'up' and 'down' (left) and the trees that score 'flat' (right).

Most of the gain, in both halves, goes to the tick size relative to the price and to volatility. They decide whether the mid moves beyond the 1 bp dead zone at all (in a low-priced stock that takes a full tick, in a high-priced one it happens constantly), and every tree, including the 'up' and 'down' trees, has to separate a move from no move. The order-book features that say *which way* (microprice tilt, queue imbalance, imbalance × one-tick share, imbalance changes, order flow) rank next in the up/down trees and barely appear in the flat trees. The new h30s_flow features take 50% of the total gain (52% in the up/down trees, 49% in the flat trees). Gain shows what the trees split on, not how much each input adds out of sample; the ablation in 4.1 is the cleaner measure of contribution.

4.3 Single-feature signal on the test sessions

Feature	IC (pooled)	IC (per session)	Set
microprice_bps	+0.089	+0.091	basic
microprice_ticks	+0.089	+0.091	h30s_flow
imbalance	+0.089	+0.091	basic
imb_x_one_tick	+0.076	+0.082	h30s_flow
imb_change_6	+0.066	+0.067	h30s_flow
imb_change_1	+0.053	+0.054	h30s_flow
imb_mean_6	+0.043	+0.045	h30s_flow

Feature	IC (pooled)	IC (per session)	Set
ofi_1	+0.039	+0.038	h30s_flow
ofi_2	+0.036	+0.035	h30s_flow
imb_mean_12	+0.029	+0.030	h30s_flow
ofi_6	+0.021	+0.019	h30s_flow
ofi_12	+0.015	+0.012	h30s_flow

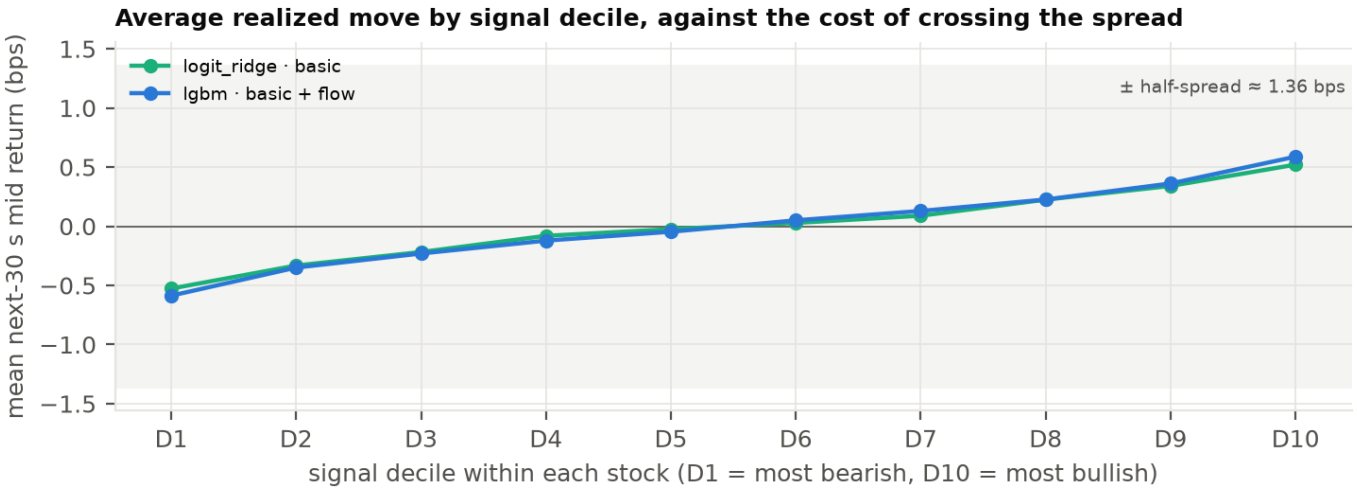
Rank correlation of each feature alone with the realized 30 s return over the test sessions (descriptive; no model). 'Pooled' ranks each stock over all test sessions and averages over stocks; 'per session' ranks within each stock-session. The two agree for the order-book features but not for the price-level ones (next paragraph).

Measurement note: per-session IC is biased for price-level features. Within one stock and one session, tick_bps (one tick ÷ price) and spread_bps (spread ÷ price, with the spread fixed at one tick) move only with the price level, and ranking a level within a single session uses that session's future prices. On a simulated random walk, where nothing is predictable, the per-session IC of 1/price with the next 30 s return averages +0.063. On our data tick_bps scores +0.077 per session but +0.005 pooled. The framework's headline IC ranks all test rows together and is not affected; the per-stock and time-of-day ICs in this report are pooled over all test sessions for the same reason (LightGBM per-stock IC: 0.103 pooled vs 0.104 per stock-session, so the model's predictability is not an artifact of this effect). Worth keeping in mind for any group that evaluates per session.

4.4 Why it works

The mid moves when one side of the best quote is used up. In a low-priced stock one cent is a large step, so the bid and ask queues are long, the spread sits at one tick and the price can only move a tick at a time; the relative size of the two queues (and how fast they grow or shrink) then says which side will break first, often many seconds ahead. In a high-priced stock a cent is a tiny step, the book refreshes many times a second and any information in the queues is used up long before 30 seconds; the published evidence (Aït-Sahalia et al., NBER w30366) likewise finds that predictability falls with the share price and sits mostly in the last milliseconds.

5. Is it economically meaningful?



Sorting each stock's predictions into deciles, the realized 30 s move rises monotonically from -0.59 bps (D1) to +0.59 bps (D10) for LightGBM (-0.53 to +0.52 for the starting model). Even the extreme deciles stay inside the ±1.36 bps half-spread band.

The model ranks future moves reliably, but the expected size of a move is smaller than the cost of crossing the spread, and R^2 against a zero forecast is under 1%. As a stand-alone taker strategy it would lose money; its value is in deciding when to post or cross, i.e. execution and market making, and as an input to longer-horizon models. The framework's economic test (*hftf ml econ*) would quantify this and has not been run yet.

6. No look-ahead bias: safeguards

Stage	Safeguard	Verified by
Universe	Fixed list of 28 large caps chosen before looking at results; not selected on performance. (The framework also supports point-in-time S&P 500 membership.)	symbols_pilot28.txt
Bars	Bars are labelled at the bar end; a bar at t contains only quotes up to t.	framework bar schema
Labels	The target shift is taken within a session, so a label never reaches into the next day; labels crossing the open/close, gaps and stale targets are dropped.	tests/unit/ml/test_labels.py
Features	Every feature is a non-negative shift or a trailing window inside the session.	check_feature_lookahead (recomputes features after deleting later bars) for basic and h30s_flow; hand-computed OFI test
Cross-validation	Walk-forward by whole sessions; training rows whose label window overlaps the test window are purged, plus an embargo after it.	tests/unit/ml/test_cv.py
Model selection	LightGBM early stopping uses held-out blocks inside the training window only.	tests/unit/ml/test_h30s_flow_lgbm.py
Data loading	load_bars bug fixed before any reported run: each bar enters the dataset exactly once, so folds and labels see the true bar sequence.	regression test in test_datasets.py
Diagnostics	The feature-importance refit uses the last fold's training rows only.	this script
Evaluation	Descriptive ICs in this report pool each stock over all test sessions: ranking within one session would let price-level features borrow that session's future prices (measurement note, 4.3).	random-walk check: per-session bias +0.063

Latest test run: **367 passed, 5 skipped** (ruff and import-linter clean; mypy shows the 20 errors that already exist on main).

7. Limitations and next steps

- **Trade flow is missing.** Signed volume and trade intensity were the next-strongest signals in our separate research on 2025 data, but the shared dataset builder loads only quote bars. Joining the trade bars we already pulled is a shared-code change (cross-group PR).
- **One dead zone for all stocks.** The 1 bp flat band is 1–2 ticks for AAPL but a fraction of a tick for F (where 70%+ of rows are flat); a spread- or tick-relative band would make classes comparable.
- **Universe and period.** 28 names over six months of 2024; BRK.B and GOOGL need a *pull-taq* fix for suffixed symbols.
- **Multiple testing.** Every evaluated configuration is a trial of experiment h30s (the deflated Sharpe divides by the count). So far: 4 completed h30s runs (3 on TAQ data, 1 synthetic smoke test) and 6 aborted attempts that produced no results.
- **Economic test.** Run *hftf ml econ* on the LightGBM run to measure what survives costs.
- **Model capacity.** The LightGBM direction booster used its whole 500-tree budget without early stopping triggering; a larger budget is a cheap check (one more trial).

Known framework issues worth a team fix: (1) *build_dataset* reuses a feature folder left half-built by a crashed run (delete *data/features/<hash>* after a failure); (2) on Windows set *PGPASSFILE=%APPDATA%\postgresql\pgpass.conf* before pulling.

Reproduce

hftf data pull-taq --manifest pilot28_2024h1_5s --start 2024-01-02 --end 2024-06-28 --grid 5s --symbols-file backend/hft_forecast/ml/horizons/h30s/symbols_pilot28.txt

hftf ml run backend/hft_forecast/ml/horizons/h30s/config.yaml --model logit_ridge

hftf ml run backend/hft_forecast/ml/horizons/h30s/config_flow.yaml --model logit_ridge

hftf ml run backend/hft_forecast/ml/horizons/h30s/config_flow.yaml --model lgbm

python scripts/h30s_report.py

Runs: logit_ridge · basic = 820759f5b4c7, logit_ridge · basic + flow = 84929c4261f2, lgbm · basic + flow = cf45c1c618ae. Seed 42, PYTHONHASHSEED=0.